

Experimental and numerical investigations of the texture evolution in copper wire drawing

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Abstract Polycrystalline copper wires are drawn in a single and multiple step for the equivalent area reduction (RA) of $\sim 33\%$. The single step and multiple step drawing process was simulated using a rate independent crystal plasticity with finite strain, which is implemented as a user routine in commercial finite element package ABAQUS. The texture of the copper wires were characterized by X-ray diffraction (XRD) and compared with the texture based finite element (FE) simulation predictions. Initial $\langle 100 \rangle$ fiber decreases during the drawing process and is replaced by $\langle 111 \rangle$ fiber. The $\langle 111 \rangle$ oriented grains are predominant in a single step drawing compared to a multiple step of the equivalent area reduction. The finite element analysis takes into account active crystallographic slip and orientation effects during the drawing process. Regions at the interface of die-wire exhibited complex textures, which was widely seen in the multiple step drawing pattern.

1 Introduction

Wire bonding is a widely used integrated circuit (IC) packaging technology in the industry because of its low cost [1, 2]. The cost is much lower compared to wafer-level packaging, tape automated bonding and flip-chip, which are also reasonably used. The wire bonding technology has many challenges to address such as rise in cost, bond pad pitch, and raw material cost of the wire. The rising cost of the gold wire, which is widely used raw material for wire bonding

applications, has turned the attention toward low cost copper (Cu) wire [3]. The price of copper has been estimated to be 10–40% of the gold (Au) and it is not subject to sudden market fluctuations. Copper wire is much cheaper compared to gold wire in the present scenario considering the volume used in the IC chip fabrication. The current market price of gold wire, diameter 20 μm 500 m spool, is approximately 200 USD, which is 1000% higher than the comparable copper wire [4]. The selection of copper stands vindicated not only based on cost alone, but owing to specific mechanical properties which are found superior to gold.

The copper wires have better electrical and thermal properties than gold wires. Copper is approximately 25% more conductive than gold, accounting for increased power rating and better heat dissipation. Higher electrical conductivity results in lowering the IC delay and less power loss [5]. Copper wires have higher tensile strength, lower wire sag, and better loop stability is obtained during encapsulation [6, 7]. Copper wires are found to have excellent ball neck strength after the ball formation [8]. Compared to gold wires, the higher stiffness of copper wires is more suitable to fine pitch bonding, leading to better looping control and less wire sagging for ultrafine-pitch wire bonding [2]. Using copper wire can be a solution to the wire short problem caused by small wire sizes, besides other solutions such as using insulated wire and having varying loop heights. High stiffness and high loop stability of Cu wire lead to better wire sweep performance during encapsulation or molding for fine-pitch devices, and can help to achieve longer/lower loop profiles [3]. The manufacturing of these polycrystalline wires requires an understanding of the relation of the role of deformation induced anisotropy on the plastic flow behavior of the material. In polycrystalline materials, the crystallographic orientations of individual grains, play an important role in the plasticity of the material [9, 10]. In face

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centered cubic material (FCC), i.e., silver, gold, copper, and brass undergoing rolling or extrusion, the grains rotate toward a preferential orientation, leading to deformation induced crystallographic texture. These texture studies have revealed orientation dependent mechanical properties [11–14]. The quantitative characterization of the texture is critically important since textures induced by the forming operations affects the subsequent processing operations because of anisotropic flow properties.

The copper wires used are produced by the cold wire drawing process. Cold drawing involves plastic cold working in which material undergoes large plastic deformation at temperatures less than $0.3T_m$ (T_m is absolute melting temperature) in several stages. Homogeneity of texture and microstructure obtained during the drawing process is a factor that affects the mechanical properties of the bonding wire. The stress inhomogeneity of a drawn wire depends on die angle α_d , area reduction (RA) and friction μ at the die–material interface. The microstructure and anisotropy of drawn polycrystalline ETP copper wire was investigated by Rajan and Petkie [15]. The texture of Cu wire had a mixture of $\langle 111 \rangle$ and $\langle 100 \rangle$ fiber components parallel to axis direction. The ratio of $\langle 111 \rangle$ to $\langle 100 \rangle$ with varying drawing strain have been reported by Inakazu et al. [16] where the fiber texture formation and the mechanical properties were analyzed. The stacking fault energy of FCC materials on the crystallographic texture was measured by English and Chin [17]. Their results showed that, low stacking fault energy material such as silver had $\sim 90\%$ $\langle 100 \rangle$ fiber texture. In this context, Stout et al. [18] observed that the initial texture strongly influences the final outcome. In copper and copper alloys, the intensity of $\langle 100 \rangle$ fiber texture decreases with increasing strain as reported by Hibbard [19]. The results showed that $\langle 111 \rangle$ is a stable texture in FCC materials [20].

The experimental texture studies which revealed inhomogeneity of the microstructure based on drawing strain paved the way for modeling the same using crystal plasticity based models. Cold drawn gold bonding wire was studied by Cho et al. [21]. The drawn wire texture was experimentally studied and compared with a simulation which applied a rate dependent crystal plasticity material model, i.e., all slip systems (both active and inactive) always slip at a rate which depends on the current stress state and slip system deformation resistance. The interaction of dislocations was neglected. Mathur and Dawson also [22] investigated the surface textures of the wire using a rate dependent crystal plasticity model.

The grain orientation depicts the characteristic anisotropic feature of a material. Mechanical properties are estimated based on this anisotropic response. The anisotropic texture is based on the cumulative plastic shear strain, which is obtained from selective active slip systems. Thus, plastic deformation is largely heterogeneous for polycrystals.

Therefore, it is worthwhile to study the deformation process using a rate independent crystal plasticity where only active slip systems are selected for the plastic flow and texture evolution, which is the subject of this paper. In this study, the evolution of texture and microstructure in cold drawn copper bonding wire was analyzed with X-ray diffraction as a function of drawing strain. The effect of equivalent area reduction (RA) by two drawing schemes, i.e., a single step and multiple step on the crystallographic texture was analyzed. A rate independent 3D crystal plasticity model incorporating finite strain theory employed in commercial code ABAQUS is applied to simulate the drawing process and study the anisotropic response in the polycrystalline copper wire. The finite element (FE) simulation was used to understand the deformation during the two drawing schemes with process parameters similar to experimental conditions. The experimental and simulated crystallographic textures of the drawn polycrystalline copper wire by the two drawing schemes are compared.

This paper is structured in the following way: The experimental methodology for wire drawing, texture prediction is discussed in Sect. 2. In Sect. 3, numerical modeling summarizes the constitutive behavior of copper and its implementation in the ABAQUS finite element program. The experimental measurements and numerical predictions are discussed in Sect. 4. Concluding remarks are given in Sect. 5.

2 Experimental procedures

2.1 Materials and methods

The copper wire of initial diameter $\varnothing$ 1.6 mm was bought from a supplier. The purity of the as received copper wire was characterized using a energy dispersive spectroscopy (EDS) and found to be more than 99.99%. The original copper wire of $\varnothing$ 1.6 mm was drawn to $\varnothing$ 1.3 mm in two drawing schemes, i.e., a single step and multiple step for the area reduction (RA) of $\sim 33\%$. In the single step drawing, the original wire of $\varnothing$ 1.6 mm was drawn to $\varnothing$ 1.3 mm in a single draw whereas in the multiple step, the wire was reduced to $\varnothing$ 1.5 mm, $\varnothing$ 1.4 mm, and then finally to $\varnothing$ 1.3 mm. The wire was drawn through tungsten carbide dies attached in a drawing plate in order to achieve the area reduction. The die angle α_d and the bearing length of the drawing die are $4.5^\circ \pm 0.5$ and 1.4 ± 0.2 mm, respectively, as specified by the die manufacturer. A solid lubricant of graphite is used to minimize the effect of friction μ during the drawing process. A controlled drawing velocity of 25 mm/s was used and the strain rate was kept constant for both the drawing schemes. The total drawing strain was ~ 0.40 from the initial as received wire to the final wire.

2.2 X-ray diffraction measurement

Experimental measurements of the texture of the as received copper wire and drawn copper wire were obtained by X-ray diffraction method with $\text{CuK}\alpha$ radiation using a Bruker D8 diffractometer. Incomplete pole figures were generated on $\{111\}$, $\{200\}$, $\{220\}$, and $\{311\}$ crystallographic planes. The irradiated surfaces were measured along the longitudinal section of the wire and the area was $\sim 0.8 \times 2.5$ mm. The average grain diameter of the as received copper wire was about 60 μm , a typical irradiated surface samples about 764 grain orientations. The incomplete pole figures in its raw form is uncorrected and is in the form of discretized intensities as a function of goniometer position angles. The raw data was processed using MULTEX, a built-in package with Bruker D8 X-ray diffraction. Each measured pole figure was corrected for background and defocussing. All pole figures are equal area projections of the specified crystallographic planes. In order to obtain complete information of a texture, orientation distribution function (ODF) is considered for the representation since a large amount of technically relevant information required for qualitative analysis of textures can be obtained from such a representation. ODF was calculated from the experimental pole figures using orthorhombic sample and cubic crystal symmetry. The symmetry requires the elementary three-dimensional Euler space defined by $0^\circ \leq \varphi_1 \leq 360^\circ$, $0^\circ \leq \phi \leq 90^\circ$ and $0^\circ \leq \varphi_2 \leq 90^\circ$.

3 Numerical modeling

Kinematics of single crystal plasticity deformation is based on finite strain crystal plasticity theory of Pierce et al. [23, 24]. The algorithm of solving crystal plasticity with finite element method and solution approach for the yield function is discussed in this section.

3.1 Constitutive model for copper single crystal

The plastic deformation in a face centered cubic (FCC) copper (Cu) crystal considered here is confined to preferred (dense) crystal planes and directions, known as slip systems α . The FCC lattice deforms on the slip systems defined by the $\{111\}$ family of slip planes m^α in the $\langle 110 \rangle$ family of slip directions s^α . There are twelve combinations of slip systems which govern the macroscopic plastic deformation of a single crystal.

The total strain $\dot{\varepsilon}$ can be divided into elastic and plastic components and is related to the deformation gradient F as follows:

$$\dot{\varepsilon} = \dot{\varepsilon}^e + \dot{\varepsilon}^p \quad (1)$$

$$\varepsilon = \frac{1}{2} [F^T F - I] \quad (2)$$

where I is the identity tensor. The deformation gradient F is calculated for each load step or increment by the ABAQUS FE program. A trial strain as in (3) is calculated assuming plastic strain $\varepsilon^p = 0$ before yield occurs. Forward Euler integration is used with small incremental time steps to solve for the plastic strain increments in each step.

$$\Delta \varepsilon_{n+1} = \Delta \varepsilon_{n+1}^e + \Delta \varepsilon_{n+1}^p \quad (3)$$

The external stress increment $\Delta \sigma$ for the corresponding trial strain increment is calculated using the

$$\Delta \sigma_{n+1} = E^e : \Delta \varepsilon_{n+1} \quad (4)$$

A cubic elastic anisotropic tensor E^e is used for the computation of the external stress applied on a single crystal in the elastic regime. The moduli values used in this simulation are $E_{11} = 170$ GPa, $E_{12} = 123.4$ GPa, and $E_{44} = 75.4$ GPa [25].

When a single crystal is subjected to an external stress σ , the resolution of this stress on to the preferred slip system is called the Schmid stress τ . When the Schmid stress is higher than the slip system resistance to yield, plastic deformation takes place. The Schmid stress increment $\Delta \tau$ on a specific slip system α is given by:

$$\Delta \tau_{n+1}^\alpha = \Delta \sigma_{n+1} : P_n^\alpha \quad (5)$$

where P^α is a symmetric space spanned by slip direction s^α and slip plane normal m^α is as:

$$P^\alpha = \frac{1}{2} (s^\alpha \otimes m^\alpha + m^\alpha \otimes s^\alpha) \quad (6)$$

The slip direction and normal are subjected to coordinate transformation at the beginning of each incremental step.

The yield criteria as defined by the following equation (7) is validated for each of the slip system.

$$\Phi_{n+1}^\alpha = (\tau_n^\alpha + \Delta \tau_{n+1}^\alpha) - \tau_0^\alpha - g_{n+1}^\alpha \quad (7)$$

The hardening variable, $g^\alpha = \tau_0$, is the initial critically resolved shear stress of the slip system. The flow rule governing the plastic strain increment is given by the (8). The plastic strain increment is computed based on the shear strain γ of each active slip system $\alpha \in A$, *activated*. A rate independent crystal plasticity case is considered here where the determination of active slip systems is based on Kuhn–Tucker criteria [26] as defined by (9), (10), and (11). The slip systems with shear strain $\gamma^\alpha < 0$ are not considered for the yield function validation during that particular time step and are excluded. The first incremental step assumes $A = 0$, i.e., no slip systems are active.

$$\Delta \varepsilon_{n+1}^p = \sum_{\alpha \in A} \gamma^\alpha \frac{\tau_n^\alpha}{|\tau_n^\alpha|} P_n^\alpha \quad (8)$$

$$\gamma^\alpha \geq 0 \quad (9)$$

$$\Phi_{n+1}^\alpha \leq 0 \quad (10)$$

$$\gamma^\alpha \Phi_{n+1}^\alpha = 0 \quad (11)$$

To compare the simulated numerical behavior with experimental response of nanoindentation measurements, the interaction of dislocations are taken into account by the hardening law g^α . The anisotropic hardening behavior of copper (Cu) as given by the (12) is applied on the activated slip systems. The hardening behavior accounts for both self hardening $h_{\alpha\alpha}$ and latent hardening $h_{\alpha\beta}$ for each slip system. The parameter q representing the ratio of latent over self hardening is taken to be 1.4 from the experimental results [25]. The hardening response, which can be seen from our previous work [27] on a single crystal copper showed stage II clearly, indicating latent hardening on the activated slip systems.

$$g_{n+1}^\alpha = g_n^\alpha + \sum_{\beta \in A} h_n^{\alpha\beta} \gamma^\beta \quad (12)$$

$$h_n^{\alpha\beta} = h_n^{\alpha\alpha} [q + (1 - q) \delta^{\alpha\beta}] \quad (13)$$

3.2 Finite element analysis

For the wire fabrication, a drawing plate with several individual dies were used to achieve the area reduction. For consistency, the wire was drawn in the same direction through each die such that the axis is parallel to the subsequent drawing direction. In single step, the die imparted a total drawing strain of ~ 0.40 whereas in multiple step, controlled drawing strains of ~ 0.12 , 0.13 , and 0.14 were achieved. The final bonding wires with 1.3 mm were achieved from the drawing schemes.

The finite element modeling characterize the essential features of the deformation imposed in the wire drawing experiments. Using the symmetry of the experimental set up, an axisymmetric analysis was carried out. A rigid conical die is used to reduce the cross sectional area of a copper wire 0.8 mm in diameter and 2.5 mm long by $\sim 33\%$ using axisymmetric finite element (FE) drawing simulations as shown in Fig. 1. The FEM workpiece consisted of 2,272 four noded axisymmetric elements, with reduced integration (CAX4R) and enhanced hour glass control. The pole figures processed from the uncorrected raw data was superimposed on the FE workpiece. The initial texture of the FE model have 568 unweighted discrete grain orientations which corresponds to the experimental crystallographic texture of the as received wire, as shown in Fig. 2. A simple tension test on an aggregate of 568 unweighted grain orientations representing a polycrystalline copper material was simulated. The experimental measurements and simulation for the stress-strain response of the as received copper wire

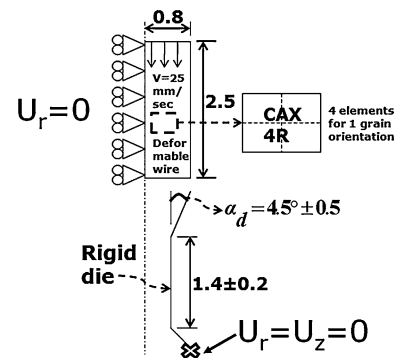


Fig. 1 Finite element schematic of wire drawing process

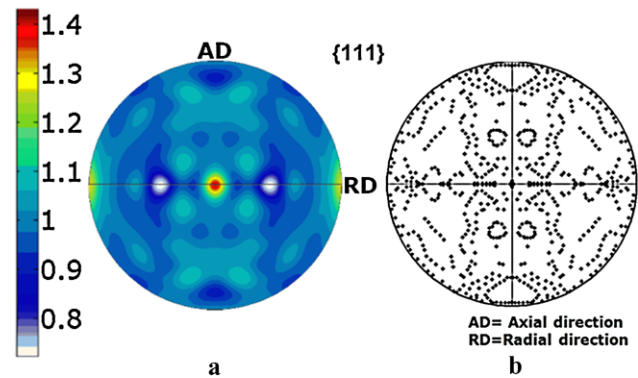


Fig. 2 Initial texture of as received copper wire: (a) experimental $\{111\}$ (equal area projection) pole figure and its (b) numerical representation by 568 grain orientations

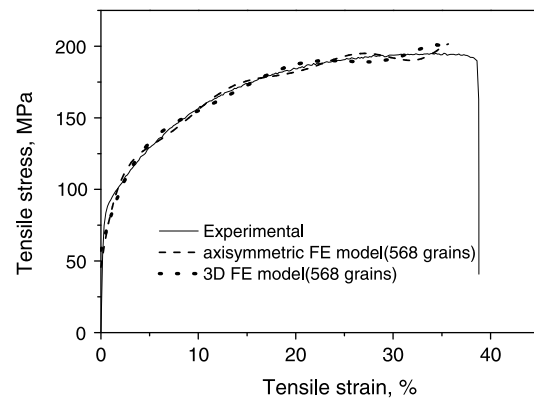


Fig. 3 Comparison of stress-strain curve in tension using 568 unweighted grain orientations with experimental measurements

are in good agreement as shown in Fig. 3. In this paper, a texture component designated by $\{hkl\} \langle uvw \rangle$ means that the $\{hkl\}$ plane normal is parallel to the radial direction (RD) and $\langle uvw \rangle$ is parallel to the drawing or axial direction (AD). With four axisymmetric elements representing each grain orientation in the finite element model, the simulation was conducted. The die angle α_d and the bearing length of the die are $4.5^\circ \pm 0.5$ and 1.4 ± 0.2 mm, respectively, which

is the same with the experimental drawing conditions. The die angles were kept the same for both the drawing schemes in the simulation. The coefficient of friction between the die and the copper wire was kept constant at a value of 0.05 for all the drawing simulations conducted. The friction in diamond die has been considered to be negligible during modeling by Cho et al. [10] and Park et al. [28, 29]. They assumed 0.001 as Coulomb coefficient of friction between the die and wire. Here, experiments were conducted in a tungsten carbide die instead of diamond, which is widely used in the industry. The friction coefficient between the tungsten carbide die and copper wire was found to be in between 0.03–0.11 as investigated by Haddi et al. [30]. Within numerical scatter, the obtained results were almost constant for friction coefficient values of 0.03, 0.05, and 0.07, and finally we selected 0.05 for all the simulations. Wire drawing is simulated by moving the wire, which is given the drawing speed of 25 mm/s causing it to traverse the die from one end to the other.

4 Results and discussion

A comprehensive assessment of the microstructural and textural evolution in the drawn copper wires by two drawing schemes is given here from the experimental measurements and its FE predictions. The microstructural inhomogeneities developed in the wire during deformation contributed substantially in understanding the fiber texture, which is discussed. The effect of drawing strain on the complex surface texture of the wire is analyzed.

4.1 Crystallographic texture

The longitudinal section of the as received wire of $\varnothing$ 1.6 mm shows predominantly a mixture of weak $\langle 111 \rangle$ and $\langle 100 \rangle$ orientations with regions of more complex textures. The texture measured by X-ray diffraction in the form of pole figure of the as received wire is shown in Fig. 2. A $\{111\}$ pole figure with AD parallel to the drawing direction shows a overall well developed $\langle 100 \rangle$ fiber texture and a random local texture. The main texture component is close to a rotated cube component $\{100\}$ $\{011\}$ or ideal Goss component. As the wire diameter decreases during drawing, equivalent plastic strain increases and strong single component crystallographic texture evolves near the drawing axis. The initial grain orientation of the polycrystalline wire also influences the crystallographic texture of the drawn wires, which is seen from the presence of $\langle 100 \rangle$ texture across the wire section in both the drawing schemes.

The pole figure of the copper wire measured and predicted using FE calculation, drawn by single step area reduction, is shown in Fig. 4. The texture of the drawn wire

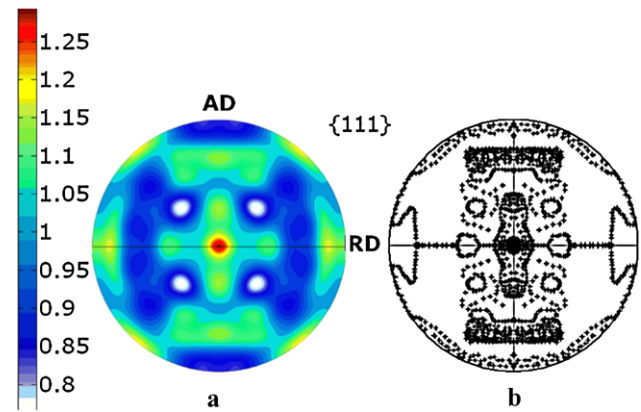


Fig. 4 Texture of drawn wire—single step: (a) experimental $\{111\}$ (equal area projection) pole figure and its (b) FE prediction

also exhibited similar duplex $\langle 111 \rangle$ and $\langle 100 \rangle$ components. However, the presence of $\langle 111 \rangle$ texture component increased in the single step drawing subjected to a area reduction of $\sim 33\%$. The presence of $\langle 100 \rangle$ component is very weak, but this property is inherited from the initial texture of the received wire. This variation correlates well with the moderately high stacking fault energy (SFE) ($\gamma = 21$ mJ/m²) of copper during deformation as reported by English and Chin [17]. The increase of $\langle 111 \rangle$ texture component is attributed to slip dependent behavior. The proportion of $\langle 100 \rangle$ texture have been argued by Calnan [31] and Bishop [32] based on latent hardening in active slip systems. The finite element prediction is based on specific slip system interaction imposed by the deformation, which is considered here. Relative amount of slip on the specific systems leads to differential hardening which is a fundamental feature in texture formation. The FE also shows a strong $\langle 111 \rangle$ texture which compares well with the experimental measurements. The $\langle 111 \rangle$ fiber texture components tends to align with the axial direction of the cold drawn wire which is clearly seen from the simulation as well as experimental measurements. A drawn wire, in the intermediate stage of a multiple step, was analyzed for the multistage drawing texture development. The experimental pole figure of the wire drawn to a strain of 0.12 and its FE prediction is shown in Fig. 5. A strong $\langle 112 \rangle$ texture parallel to the drawing direction and a random $\langle 110 \rangle$ texture is observed. The comparison with finite element simulation shows a good agreement for strong $\langle 112 \rangle$ texture component. The pole figures show a weak drawing texture after the intermediate drawing step. This shows the heterogeneous nature of plastic deformation. At low strains, Cu shear band formation was extrapolated from the pole figures. Although, the shear bands are observed in low to high strain wires, they were aligned at 60° to the AD parallel to the drawing direction in low strains (0.12), another interesting observation is that, the shear bands tend to closely relate with the Brass component $\{110\}$ $\langle 112 \rangle$ with

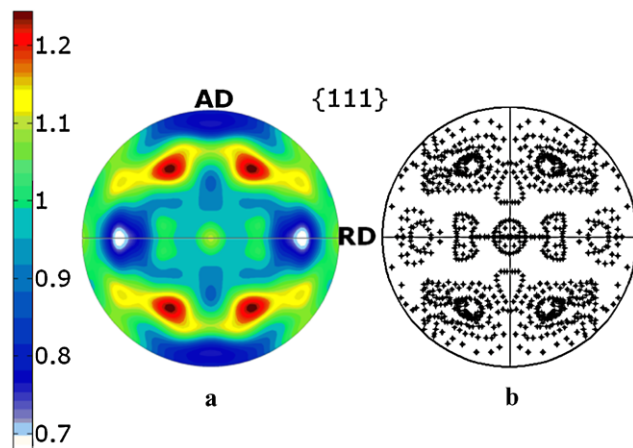


Fig. 5 Texture of drawn wire—intermediate step: (a) experimental $\{111\}$ (equal area projection) pole figure and its (b) FE prediction

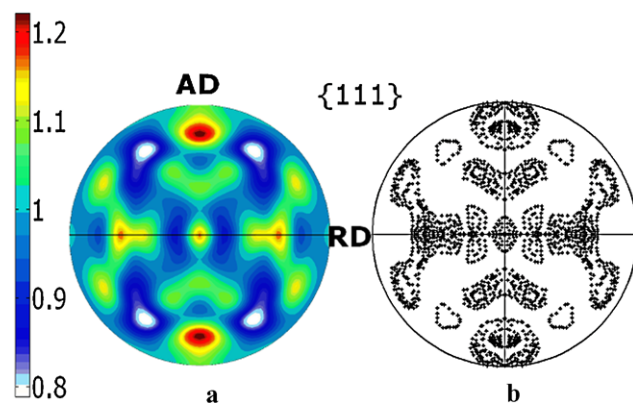


Fig. 6 Texture of drawn wire—Multiple step: (a) experimental $\{111\}$ (equal area projection) pole figure and its (b) FE prediction

a tolerance angle of 15° from the ideal orientation. The rotation of the shear bands toward the AD (drawing) direction is shown through the multiple step drawn wire texture experimental measurement and its FE predicted pole figure, as shown in Fig. 6. As the strains in the wire increase, the shear bands tend to rotate toward the $\langle 111 \rangle$ or $\langle 210 \rangle$. The $\langle 111 \rangle$ grains start to appear throughout the section as the applied strain increases in the multistage drawing process. It is inferred that, at the strain of 0.40, the $\langle 100 \rangle$ oriented grains does not change much in both the drawing schemes but the $\langle 111 \rangle$ increases monotonically in the single step drawing compared to multiple step. At high strains, the texture components are predominantly a mixture of $\langle 111 \rangle$ and $\langle 100 \rangle$ oriented grains, which is observed from both the drawing steps. The misorientation angle of the shear band in the multiple step wire lies between 10° – 15° from the drawing direction. This transformation has been observed in copper and its alloys [33, 34]. Despite the heavy deformation applied in both the drawing steps, deformation twins were not observed in the drawn

wires. Deformation twins in copper and its alloys subjected to room temperature forming operations are rarely reported. This is due to presence of ample slip systems to accommodate the plastic deformation. The complex fiber texture component is decreasing with the increase in drawing strain as observed from the two drawing schemes analyzed.

The crystallographic texture exerts significant influence on many physical and mechanical properties of the deformed materials. The analysis of fiber texture evolution in the cold drawn wire during plastic deformation helps in understanding the deformation mechanisms associated with the wire drawing. The ODF of $\langle 111 \rangle + \langle 100 \rangle$ oriented regions in the wire is systematically analyzed based on the drawing strain. The ODFs of received wire, single step, intermediate step, and multiple step drawn wires are shown in Figs. 7(a) to 7(d). The ODF is extracted from the pole figure using WIMV method [35]. For a two-dimensional representation, the Euler space is subdivided into cells or boxes. The two axes chosen here for the representation: sections of φ_1 with a cross section at $\varphi_2 = 45^\circ$.

The major texture component of the received wire as observed from the ODF in Fig. 7(a) confirms the presence of a Goss component being predominant over the whole area followed by Copper and Brass components. The overall texture of the wire shows a cylindrical symmetry with a random local texture. In the $\langle 111 \rangle + \langle 100 \rangle$ regions of the drawn wire, exhibit α -fiber (Goss–Brass) and β -fiber (Brass–S–Copper) components. The ODFs show a strong α -fiber and β -fiber components during the single step drawing compared to the multiple step drawn wire. The intermediate step drawn wire has a weak α -fiber and β -fiber components with the Brass component dominant which verifies the pole figure measurements. There are still a few Copper and Goss oriented grains as observed in the $\langle 111 \rangle + \langle 100 \rangle$ region of the intermediate step drawn wire. As the drawing deformation increases, Brass component is observed to be very weak, while the Copper component becomes prevalent in the drawn wire manufactured by single step compared to multiple step drawn wires. Although the complex texture component decreases with increase in drawing strain, the S and Brass components still exists.

From the rate independent crystal plasticity theory, the inhomogeneous deformation observed from the ODFs, is due to the latent hardening applied on the specific active slip systems of the grains. The finite element takes into account heterogeneous slip interaction during the drawing process. The shear strain in each slip system of the grain depends on the loading rate, which causes the slip reaction between the grains. Park et al. [28, 29] using a rate dependent crystal plasticity theory, assuming no hardening between the grains, have shown that the shear strain contribution to the metal flow behavior decreases with increasing drawing strain and remains unaffected. At high drawing strains of 0.40 obtained

Fig. 7 ODFs ($\varphi_2 = 45^\circ$) for $\langle 111 \rangle + \langle 100 \rangle$ fiber texture components of the wires

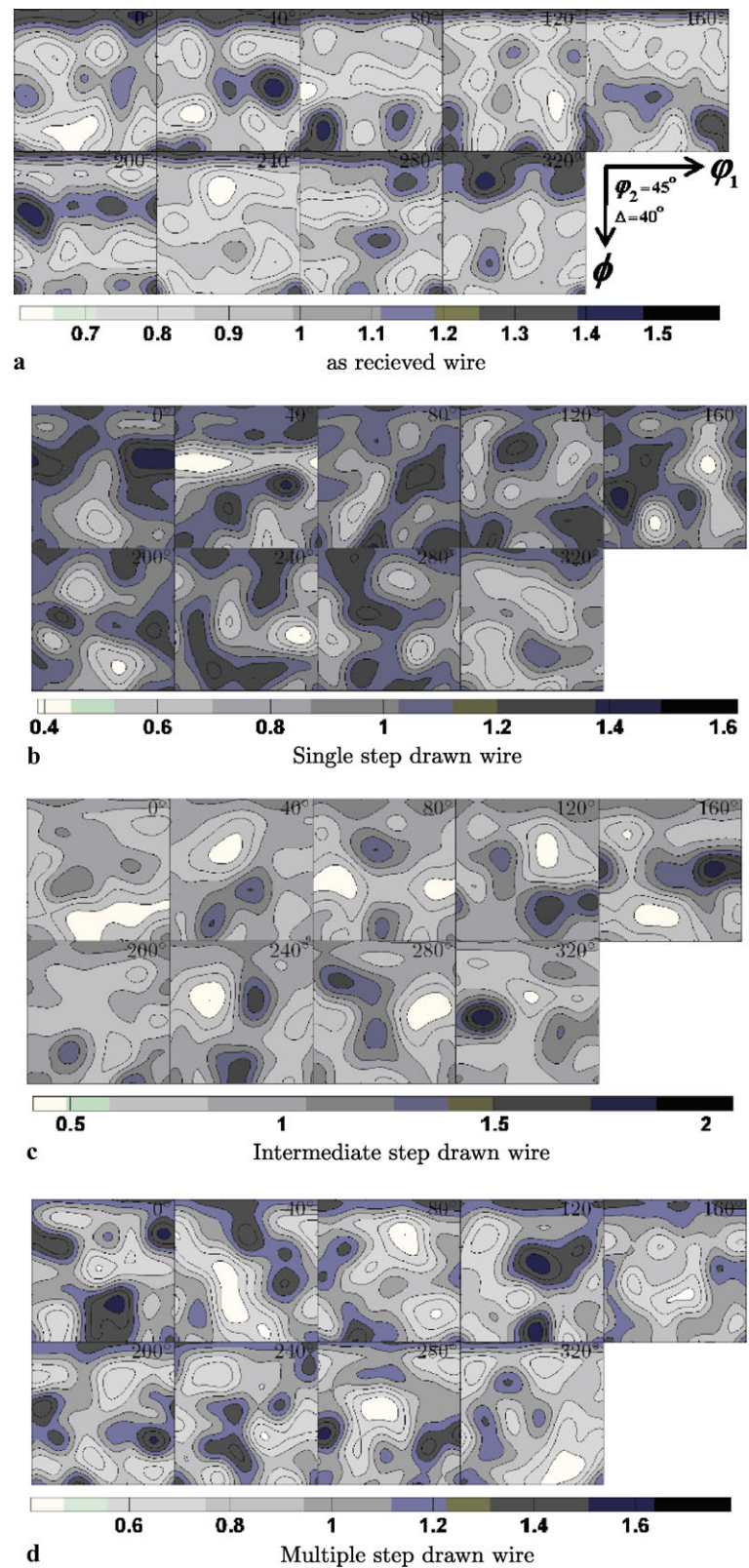
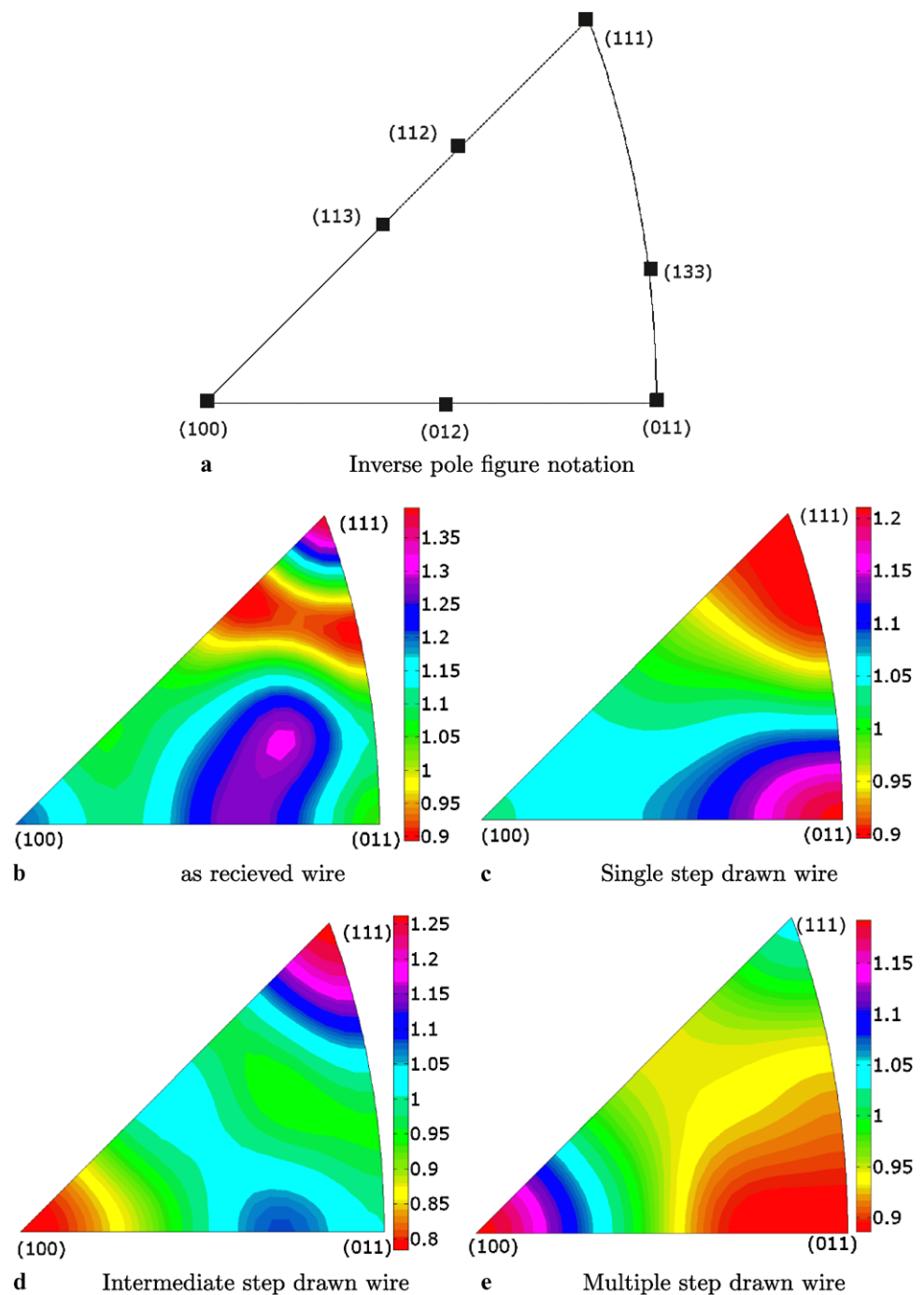


Fig. 8 Inverse pole figures of drawn wire surface texture calculated from the FE calculations



from both the drawing schemes, the shear strain contribution on the grain becomes relatively unaffected which is also verified from the ODFs based on the fiber texture evolution. The shear deformation at low strains of 0.12, plays a vital role in the latent hardening of the grains which contributes to the texture evolution.

4.2 Surface texture

Frictional tractions in metal forming process such as rolling, wire drawing, and extrusion leads to surface textures differ-

ing from those inside the workpiece, as reported in the literature. The surface texture of rolled sheets has been investigated and found to vary with the work piece predominant texture. Shear deformation due to frictional effects in the roll was observed to play a role in the sheet texture sharpening. Therefore, it is important to pay special attention to the surface texture of the drawn wire. The surface texture of the copper wire is studied and analyzed based on FE modeling. Figure 8 shows inverse pole figure (IPF) maps for longitudinal sections of the wire. The projection of wire axis onto the

(100) (011) (111) standard triangle is shown in Fig. 8(a), which is used to represent the die–wire interface texture. In this study, a minimum friction is applied on the die–wire interface. Shear deformation on the surface of the wire due to workpiece–tool frictional contact is not analyzed in this study. Mathur and Dawson [22] have pointed out that the influence of friction on surface texture of the drawn wire is negligible. In the drawing schemes studied with varying deformation strain applied, shear deformation on the surface was found to have contributed in the textural development.

The IPF depicting the surface texture of the received wire, single, intermediate, and multiple step drawn wires is shown in Figs. 8(b)–8(e). In the surface, the shear strain increases and deformation deviates from ideal drawing condition. The texture of the surface shows a decrease in $\langle 111 \rangle + \langle 100 \rangle$ components and an increase in complex texture components. The surface texture of the received wire shows a deviation from the drawing texture, which is also observed from the single, intermediate, and multiple step drawn wires. The $\langle 111 \rangle + \langle 100 \rangle$ texture components is seen to appear marginally on the wire surface in all drawing conditions. This may be related to the minimum friction in the die–wire interface. The random texture regions are apparent in the received wire as seen in Fig. 8(b). The single step drawing shows a homogeneous surface texture as seen in Fig. 8(c). The intensity of the complex regions increases in the intermediate step drawn wire as shown in Fig. 8(d) compared to the multiple step drawn wire as in Fig. 8(e). The complex texture components is related to the shear deformation of the wire surface.

The volume fraction for the complex texture components of the received and drawn wires is shown in Fig. 9. The initial texture of the received wire were random with a large volume fraction for complex textures as seen in Fig. 9(a). The Brass, S, and Copper components increases in the center of the wire as the deformation strain. However, they were seen to decrease on the surface of the wire. The volume fraction for complex textures decreases in single step drawn wire surface, which can be seen in Fig. 9(b). The $\{112\}\langle 110 \rangle$ and $\{111\}\langle 112 \rangle$ texture volume fraction tend to increase in the complex texture region. The $\{112\}\langle 110 \rangle$ complex texture component increases significantly in the single step drawn wire compared to the received wire. The volume fraction for the $\{112\}\langle 110 \rangle$ complex texture component remains unaffected in the intermediate step, however, the $\{110\}\langle 112 \rangle$ oriented texture components increases as shown in Fig. 9(c). The multiple step drawn wire surface shows a decrease in $\{110\}\langle 112 \rangle$ component, however, the $\{112\}\langle 110 \rangle$ complex texture component increases as compared to the intermediate step wire as shown in Fig. 9(d). This agrees well with the results of Rajan and Petkie [15] on the drawn copper wire surface texture.

The shear strain plays a vital role in the wire–die interface texture evolution. The shear strain at the wire surface

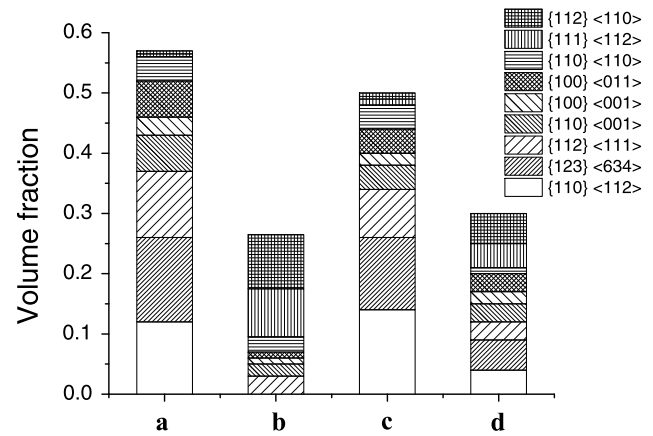


Fig. 9 Volume fraction for complex texture components of (a) as received wire, (b) single step drawn wire, (c) intermediate step drawn wire, and (d) multiple step drawn wire

during the single step drawing is lower compared to the intermediate and multiple step. The low shear strain at the surface leads to activation of all the available slip systems, thus leads to a more homogeneous deformation. The latent hardening on the surface grain ignores the strain hardening flow rule because of a negligible shear in the surface. In the intermediate step, the shear strain increases on the surface and the latent hardening on the individual grains is heterogeneous with selectively activated slip systems. The uneven shear strain leads to an inhomogeneous distribution of texture components on the wire surface, which has also been observed by Cho et al. [21]. The processing strain during the multiple drawing step increase as the shear strain on the wire surface reduces, which tend to have negligible influence on the texture evolution, thereby, the deformation is fairly homogeneous.

5 Conclusions

High purity polycrystalline Cu wire was drawn in a single and multiple step for the equivalent area reduction (RA) of $\sim 33\%$. The drawn wires microstructure was characterized by XRD and analyzed as a function of drawing strain. In order to understand the effect of deformation process on texture evolution, the drawing process is numerically simulated using a rate independent finite crystal plasticity model, which is implemented as a user routine in a commercial finite element package ABAQUS. The following major observations are made:

1. The pole figure of the received wire across the longitudinal cross section had a $\langle 100 \rangle$ texture parallel to the drawing direction and a weak random local texture. As the drawing strain increased to ~ 0.40 in the single step, $\langle 111 \rangle$ texture component increased in strength. The intermediate step drawn wire subjected to strain of ~ 0.12

exhibited strong $\langle 112 \rangle$ texture components and a weak $\langle 110 \rangle$ texture. The $\langle 112 \rangle$ texture component rotated to $\langle 111 \rangle$ texture during the multiple step drawing. The $\langle 100 \rangle$ texture was present in the cross section during both the drawing schemes. The complex texture component decreases with the increase in drawing strain. The $\langle 111 \rangle$ texture component is observed to be unstable at low drawing strains (see Figs. 4–6).

2. The ODF of the received wire confirmed the presence of ideal Goss component being dominant with Copper and Brass components. The single step drawn wire exhibited strong α -fiber and β -fiber components while the Brass component was prevalent in the intermediate step drawn wire. The Copper and Goss components remnant from the received wire was also present. As the drawing strain increases, the Brass components becomes weak and a strong Copper component was observed in the multiple step drawn wire. The strength of the Copper component was relatively less compared to single step drawn wire, (see Figs. 7(a)–7(d)).
3. The IPF representing the surface texture was analyzed based on shear deformation during the drawing process, (see Fig. 8). The surface texture of the wire deviated from the ideal drawing texture due to the the shear deformation playing a major role along the radial cross section of the wire. Complex texture components were seen on the surface of the received wire. During the single step drawing, the surface texture had an increase in strength of the $\langle 110 \rangle$ oriented grains. The $\langle 112 \rangle$ texture component strength increased on the surface during the intermediate step with complex texture prevalent. In the multiple step drawn wire, the complex texture strength reduced and $\langle 100 \rangle$ oriented grains appeared marginally.

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